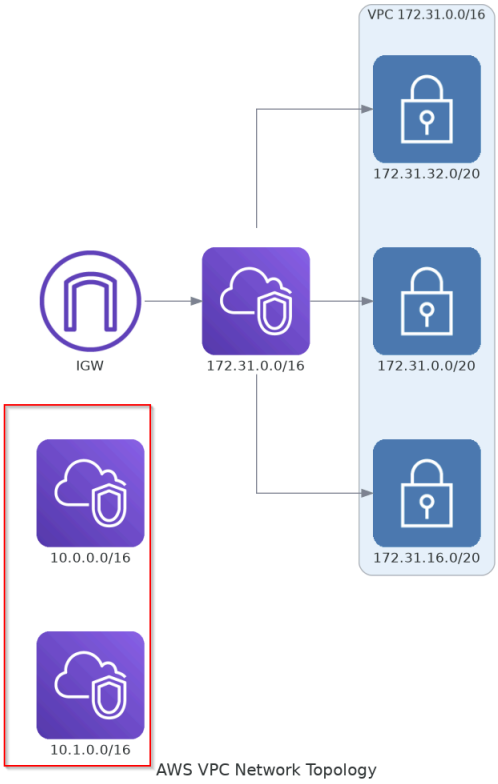
Terraform Modules Lab – Deploying VPC and S3 for Dev & Prod



Sommaire

Terraform Modules Lab – Deploying VPC and S3 for Dev & Prod..... 1

Sommaire..... 1

1 Objectifs..... 2

2 Requis..... 2

3 Structure du projet..... 2

4 Module VPC..... 3

4.1 modules/vpc/variables.tf..... 3

4.2 modules/vpc/main.tf.....3
4.3 modules/vpc/outputs.tf.....4
5 Module S3.....4
5.1 modules/s3/variables.tf.....4
5.2 modules/s3/main.tf.....4
5.3 modules/s3/outputs.tf.....4
6 Root module.....4
6.1 main.tf.....4
6.2 variables.tf (optionnel).....5
6.3 outputs.tf.....6
7 Exécution.....6
8 Résultats attendus.....10
9 Vérifications.....11
9.1 terraform output.....11

```
PS C:\terraform\terraform-vpc-s3-modules> terraform output
s3_dev_bucket = "my-dev-bucket-hguillot-2026"
s3_prod_bucket = "my-prod-bucket-hguillot-2026"
vpc_dev_id = "vpc-079d760715a26770e"
vpc_prod_id = "vpc-0c04fa78cc215ea0b"
PS C:\terraform\terraform-vpc-s3-modules>
PS C:\terraform\terraform-vpc-s3-modules>
```

.....11
9.2 terraform state list.....11
9.3 VPC.....11
9.3 S3.....12
10 éléments de la topologie crées. :.....13

1 Objectifs

Faire un simple Terraform avec vpc, un de prod et un de dev, pas de ec2, pas de Dynamodb, ni rds. Ajouter un s3 dans chaque vpc. Donc création de 4 ressources, : 2s3, 2vpc. Le faire avec des modules

2 Requis

Installer terraform

Installer AWS CLI

Avoir un compte AWS

Configurer les credentials (aws configure)

renseigner : Access Key , Secret Key, region (ex: ca-central-1) , output format (json)

3 Structure du projet

```
terraform-vpc-s3-modules/
│
├── main.tf
├── variables.tf
├── outputs.tf
│
├── modules/
│   ├── vpc/
│   │   ├── main.tf
│   │   ├── variables.tf
│   │   └── outputs.tf
│   │
│   └── s3/
│       ├── main.tf
│       ├── variables.tf
│       └── outputs.tf
```

```
PS C:\terraform\terraform-vpc-s3-modules> dir


    Directory: C:\terraform\terraform-vpc-s3-modules


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         3/28/2026  12:37 PM                modules
-a----         3/28/2026  12:55 PM            543 main.tf
-a----         3/28/2026  12:56 PM            260 outputs.tf


PS C:\terraform\terraform-vpc-s3-modules>
```

4 Module VPC

4.1 modules/vpc/variables.tf

```
variable "cidr_block" {
  type = string
}

variable "name" {
  type = string
}
```

4.2 modules/vpc/main.tf

```
resource "aws_vpc" "this" {
  cidr_block = var.cidr_block

  tags = {
    Name = var.name
  }
}
```

4.3 modules/vpc/outputs.tf

```
output "vpc_id" {
  value = aws_vpc.this.id
}
```

5 Module S3

5.1 modules/s3/variables.tf

```
variable "bucket_name" {
  type = string
}
```

5.2 modules/s3/main.tf

```
resource "aws_s3_bucket" "this" {
  bucket = var.bucket_name

  tags = {
    Name = var.bucket_name
  }
}
```

5.3 modules/s3/outputs.tf

```
output "bucket_name" {
  value = aws_s3_bucket.this.id
}
```

6 Root module

6.1 main.tf

```
provider "aws" {
  region = "ca-central-1"
}

# 🔹 AWS Provider
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }

  required_version = ">= 1.3"
}

# 🔹 VPC DEV
module "vpc_dev" {
  source     = "./modules/vpc"
  cidr_block = "10.0.0.0/16"
  name       = "vpc-dev"
}

# 🔹 VPC PROD
module "vpc_prod" {
```

```
  source     = "./modules/vpc"
  cidr_block = "10.1.0.0/16"
  name       = "vpc-prod"
}

# 🔹 S3 DEV
module "s3_dev" {
  source     = "./modules/s3"
  bucket_name = "my-dev-bucket-hguillot-2026
"
}

# 🔹 S3 PROD
module "s3_prod" {
  source     = "./modules/s3"
  bucket_name = "my-prod-bucket-hguillot-2026
"
}
```

6.2 variables.tf (optionnel)

```
# vide pour ce lab simple
```

6.3 outputs.tf

```
output "vpc_dev_id" {
  value = module.vpc_dev.vpc_id
}

output "vpc_prod_id" {
  value = module.vpc_prod.vpc_id
}

output "s3_dev_bucket" {
  value = module.s3_dev.bucket_name
}

output "s3_prod_bucket" {
  value = module.s3_prod.bucket_name
}
```

7 Exécution

Terraform init

```
PS C:\terraform\terraform-vpc-s3-modules> terraform init
Initializing the backend...
Initializing modules...
- s3_dev in modules\s3
- s3_prod in modules\s3
- vpc_dev in modules\vpc
- vpc_prod in modules\vpc
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.38.0...
- Installed hashicorp/aws v6.38.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\terraform\terraform-vpc-s3-modules>
```

L'initialisation entraine la Création du dossier .terraform et du fichier de lock . terraform.lock.hcl

```
PS C:\terraform\terraform-vpc-s3-modules> dir


    Directory: C:\terraform\terraform-vpc-s3-modules


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         3/28/2026  12:58 PM                .terraform
d-----         3/28/2026  12:37 PM                modules
-a----         3/28/2026  12:58 PM           1377 .terraform.lock.hcl
-a----         3/28/2026  12:55 PM            543 main.tf
-a----         3/28/2026  12:56 PM            260 outputs.tf


PS C:\terraform\terraform-vpc-s3-modules>
```

Entraine aussi la création des deux sous dossiers module et provider dans le dossier .terraform

```
PS C:\terraform\terraform-vpc-s3-modules\.terraform> dir


    Directory: C:\terraform\terraform-vpc-s3-modules\.terraform


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         3/28/2026  12:58 PM                modules
d-----         3/28/2026  12:58 PM                providers


PS C:\terraform\terraform-vpc-s3-modules\.terraform>
```

Le répertoire module contient le fichier modules.json .

Ce répertoire est celui dans lequel terraform downloade les modules distants (remote)

Par contre les modules locaux y sont copiés lors de la phase d' linitiaisation (terraform init)

```
PS C:\terraform\terraform-vpc-s3-modules\.terraform> cd modules
PS C:\terraform\terraform-vpc-s3-modules\.terraform\modules> dir


    Directory: C:\terraform\terraform-vpc-s3-modules\.terraform\modules


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         3/28/2026  12:58 PM            294 modules.json


PS C:\terraform\terraform-vpc-s3-modules\.terraform\modules>
```

Contenu du fichier : modules.json :

```
PS C:\terraform\terraform-vpc-s3-modules\.terraform\modules> cat modules.json
{"Modules":[{"Key":"","Source":"","Dir":"."},{"Key":"s3_dev","Source":"./modules/s3","Dir":"modules/s3"},{"Key":"s3_prod
","Source":"./modules/s3","Dir":"modules/s3"},{"Key":"vpc_dev","Source":"./modules/vpc","Dir":"modules/vpc"},{"Key":"vpc
_prod","Source":"./modules/vpc","Dir":"modules/vpc"}]}
PS C:\terraform\terraform-vpc-s3-modules\.terraform\modules>
```

Contenu du sous-dossier providers :

LE dossier contient les plugins et modules du providers choisis. Ici : AWS

```
PS C:\terraform\terraform-vpc-s3-modules\.terraform> cd providers
PS C:\terraform\terraform-vpc-s3-modules\.terraform\providers> dir


    Directory: C:\terraform\terraform-vpc-s3-modules\.terraform\providers


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         3/28/2026  12:58 PM                registry.terraform.io


PS C:\terraform\terraform-vpc-s3-modules\.terraform\providers>
```

```
PS C:\terraform\terraform-vpc-s3-modules\.terraform\providers> cd registry.terraform.io
PS C:\terraform\terraform-vpc-s3-modules\.terraform\providers\registry.terraform.io> dir


    Directory: C:\terraform\terraform-vpc-s3-modules\.terraform\providers\registry.terraform.io


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         3/28/2026  12:58 PM                hashicorp
```

Terraform validate

Terraform plan :

```
Plan: 4 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + s3_dev_bucket  = (known after apply)
  + s3_prod_bucket = (known after apply)
  + vpc_dev_id     = (known after apply)
  + vpc_prod_id    = (known after apply)
```

Teeraform apply

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

module.vpc_prod.aws_vpc.this: Creating...
module.vpc_dev.aws_vpc.this: Creating...
module.s3_prod.aws_s3_bucket.this: Creating...
module.s3_dev.aws_s3_bucket.this: Creating...
module.s3_dev.aws_s3_bucket.this: Creation complete after 2s [id=my-dev-bucket-unique-12345]
module.vpc_prod.aws_vpc.this: Creation complete after 2s [id=vpc-0c04fa78cc215ea0b]
module.vpc_dev.aws_vpc.this: Creation complete after 2s [id=vpc-079d760715a26770e]

│ Error: creating S3 Bucket (my-prod-bucket-unique-12345): operation error S3: CreateBucket, https response error Status
Code: 409, RequestID: 0XJ3E9BNBY29TJ55, HostID: 4RkGJOp4sCsP6Km1StCYBu0GxMCwLB7CVwA2GH5IvbgLxGPfV4YbrXpHv9JBGqLrfdDapt90
hfI=, BucketAlreadyExists: The requested bucket name is not available. The bucket namespace is shared by all users of th
e system. Please select a different name and try again.
│
│   with module.s3_prod.aws_s3_bucket.this,
│   on modules\s3\main.tf line 1, in resource "aws_s3_bucket" "this":
│    1:  resource "aws_s3_bucket" "this" {
│
│

PS C:\terraform\terraform-vpc-s3-modules>
```

Erreur , car le nom du bucket S3 n'est pas unique.

my-prod-bucket-unique-12345 est un nom qui a déjà été pris par qu'elqun d'autre.

Donc on modifie le nom du bucket et on le fait au niveau du code des modules et ressources qui y font référence. Puis on recommence les commandes terraform plan, terraform apply

Nouveau nom : bucket_name = my-prod-bucket-hguillot-2026 et my-dev-bucket-hguillot-2026

Puis , on modifie le main.tf du root module :

```
# 🔹 S3 DEV
module "s3_dev" {
  source      = "./modules/s3"
  bucket_name = "my-dev-bucket-hguillot-2026

"
}

# 🔹 S3 PROD
module "s3_prod" {
  source      = "./modules/s3"
  bucket_name = "my-prod-bucket-hguillot-2026
```

on fait a nouveau le cycle terraform, init, plan, apply

<u>résultat du terraform apply : figure 1</u>

on vois que par rapport au plan précédent, une ressource a été remplacée (destroy puis recreate) et que une ressource a été crée

<u>figure 1 :</u>

```
Plan: 2 to add, 0 to change, 1 to destroy.

Changes to Outputs:
  ~ s3_dev_bucket  = "my-dev-bucket-unique-12345" -> (known after apply)
  + s3_prod_bucket = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

module.s3_dev.aws_s3_bucket.this: Destroying... [id=my-dev-bucket-unique-12345]
module.s3_prod.aws_s3_bucket.this: Creating...
module.s3_dev.aws_s3_bucket.this: Destruction complete after 0s
module.s3_dev.aws_s3_bucket.this: Creating...
module.s3_prod.aws_s3_bucket.this: Creation complete after 1s [id=my-prod-bucket-hguillot-2026]
module.s3_dev.aws_s3_bucket.this: Creation complete after 1s [id=my-dev-bucket-hguillot-2026]

Apply complete! Resources: 2 added, 0 changed, 1 destroyed.

Outputs:

s3_dev_bucket = "my-dev-bucket-hguillot-2026"
s3_prod_bucket = "my-prod-bucket-hguillot-2026"
vpc_dev_id = "vpc-079d760715a26770e"
vpc_prod_id = "vpc-0c04fa78cc215ea0b"
PS C:\terraform\terraform-vpc-s3-modules>
```

8 Résultats attendus

👉 Création de :

- 2 VPC :
  - dev
  - prod
- 2 S3 :
  - dev
  - prod

👉 Donc **4 ressources AWS au total (indiquées en figure 1)**

```
s3_dev_bucket = "my-dev-bucket-hguillot-2026"
s3_prod_bucket = "my-prod-bucket-hguillot-2026"
vpc_dev_id = "vpc-079d760715a26770e"
vpc_prod_id = "vpc-0c04fa78cc215ea0b"
```

9 Vérifications

9.1 terraform output

Taper la commande : terraform output

```
PS C:\terraform\terraform-vpc-s3-modules> terraform output
s3_dev_bucket = "my-dev-bucket-hguillot-2026"
s3_prod_bucket = "my-prod-bucket-hguillot-2026"
vpc_dev_id = "vpc-079d760715a26770e"
vpc_prod_id = "vpc-0c04fa78cc215ea0b"
PS C:\terraform\terraform-vpc-s3-modules>
PS C:\terraform\terraform-vpc-s3-modules>
```

9.2 terraform state list

Taper la commande : terraform state list

```
PS C:\terraform\terraform-vpc-s3-modules> terraform state list
module.s3_dev.aws_s3_bucket.this
module.s3_prod.aws_s3_bucket.this
module.vpc_dev.aws_vpc.this
module.vpc_prod.aws_vpc.this
PS C:\terraform\terraform-vpc-s3-modules>
```

9.3 VPC

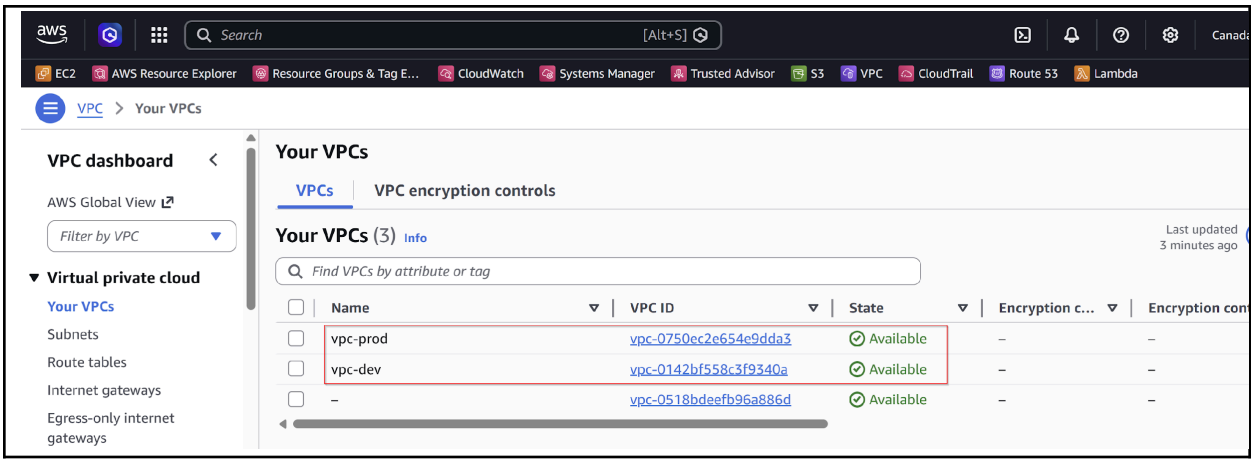
Commande CLI :

aws ec2 describe-vpcs --query "Vpcs[*].[VpcId,CidrBlock,Tags[?Key=='Name'].Value|[0]]" --output table

```
PS C:\terraform\terraform-vpc-s3-modules>
PS C:\terraform\terraform-vpc-s3-modules> aws ec2 describe-vpcs  --query "Vpcs[*].[VpcId,CidrBlock,Tags[?Key=='Name'].Va
lue|[0]]"   --output table
-----------------------------------------------------
|                    DescribeVpcs                     |
+-----------------------+----------------+-----------+
|  vpc-0750ec2e654e9dda3 |  10.1.0.0/16    |  vpc-prod |
|  vpc-0142bf558c3f9340a |  10.0.0.0/16    |  vpc-dev  |
|  vpc-0518bdeefb96a886d |  172.31.0.0/16  |  None     |
+-----------------------+----------------+-----------+

PS C:\terraform\terraform-vpc-s3-modules>
```

Dans AWS sur la console :

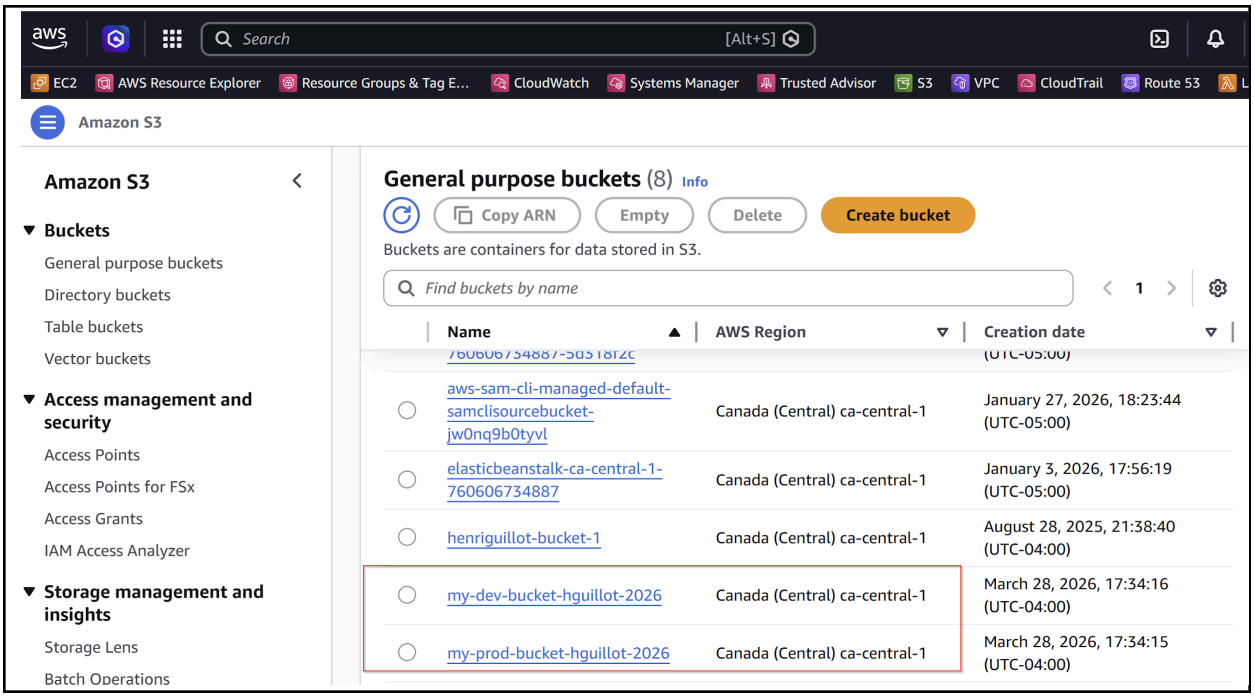


9.3 S3

Commande CLI :

```
PS C:\terraform\terraform-vpc-s3-modules>
PS C:\terraform\terraform-vpc-s3-modules> aws s3 ls
2026-01-05 13:58:05 aws-cloudtrail-logs-760606734887-5d318f2c
2026-01-27 19:38:26 aws-sam-cli-managed-default-samclisourcebucket-jw0nq9b0tyvl
2026-01-03 18:10:56 elasticbeanstalk-ca-central-1-760606734887
2025-08-28 22:01:29 henriguillot-bucket-1
2026-03-28 17:34:16 my-dev-bucket-hguillot-2026
2026-03-28 17:34:15 my-prod-bucket-hguillot-2026
2025-10-11 21:28:03 test-12
2025-10-14 17:38:36 vedge-lab-bucket
PS C:\terraform\terraform-vpc-s3-modules>
```

Dans AWS , sur la console :



---

10 éléments de la topologie crées. :

On peut vérifier la topologie crée. (la version actuelle , ne montre pas les buckets S3). Ce avec un outil de mapping graphique :

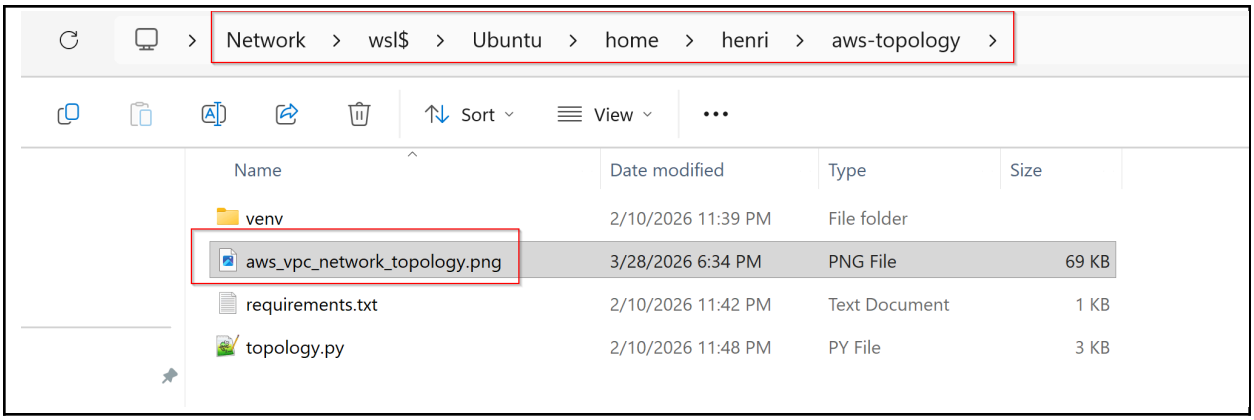
On utilise le projet : Lab topologie réseau . La référence du projet est dans github au niveau de :

On se positionne dans le répertoire terraform de ce projet :

Dans notre cas particulier ce sera :

- \\wsl$\Ubuntu\home\henri
- Puis , aller dans le répertoire : aws-topology

```
(venv) henri@Henri:~/aws-topology$ cd ..
(venv) henri@Henri:~$ dir
aws-topology  containers.json  vmimport-role-policy.json   vyos.iso
cloudmapper   venv             vmimport-trust-policy.json  vyos.qcow2
(venv) henri@Henri:~$
(venv) henri@Henri:~$ pwd
/home/henri
(venv) henri@Henri:~$
(venv) henri@Henri:~$
```

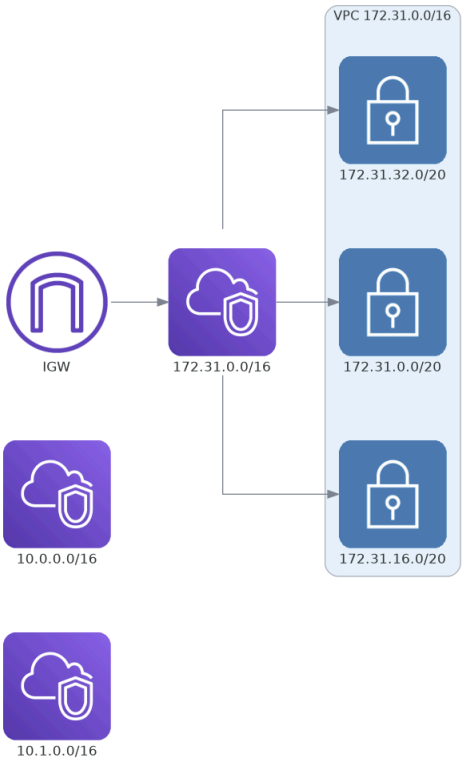


On exécute les commandes :

wsl

```
cd ~/aws-topology
source venv/bin/activate
python topology.py
```

le résultat est dans le fichier .png suivant : aws_vpc_network_topology



AWS VPC Network Topology